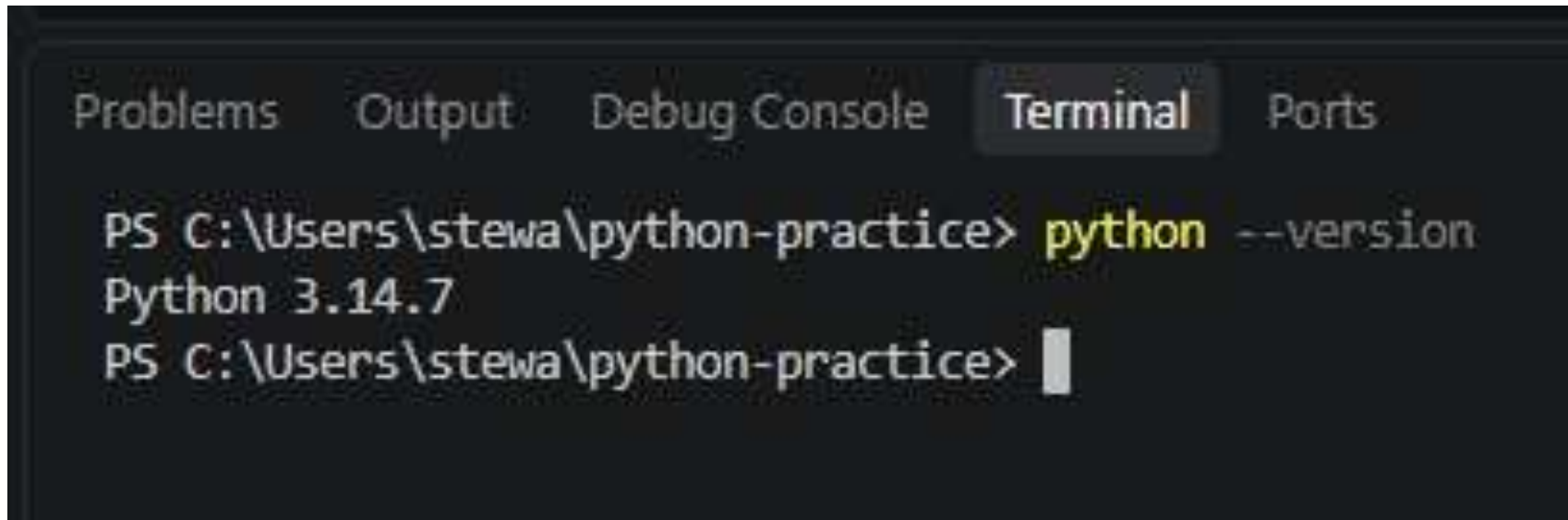


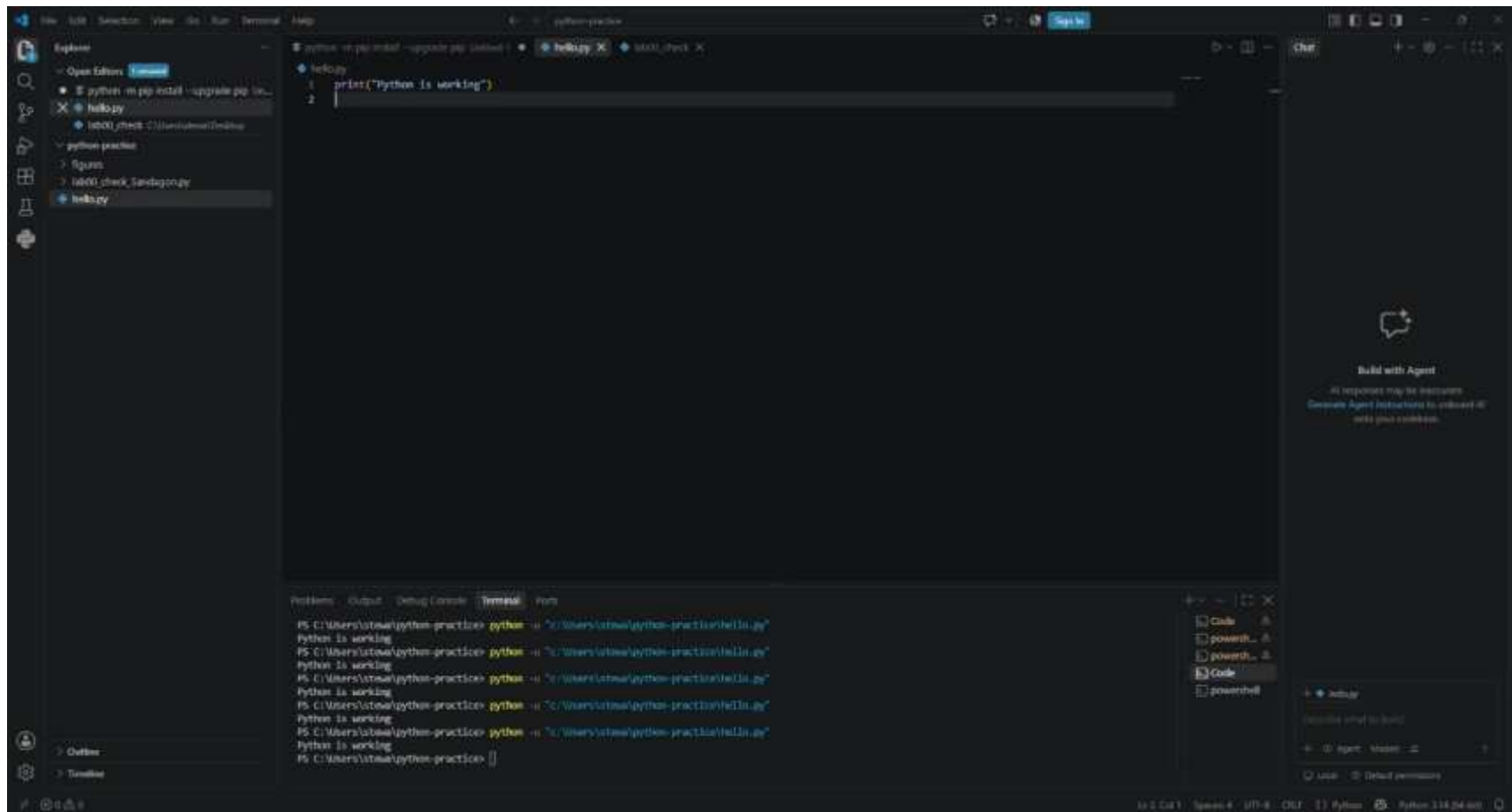
1. The terminal showing `python --version` and its answer.



```
Problems  Output  Debug Console  Terminal  Ports

PS C:\Users\stewa\python-practice> python --version
Python 3.14.7
PS C:\Users\stewa\python-practice> |
```

2. VS Code with `hello.py` open and its output visible in the terminal.

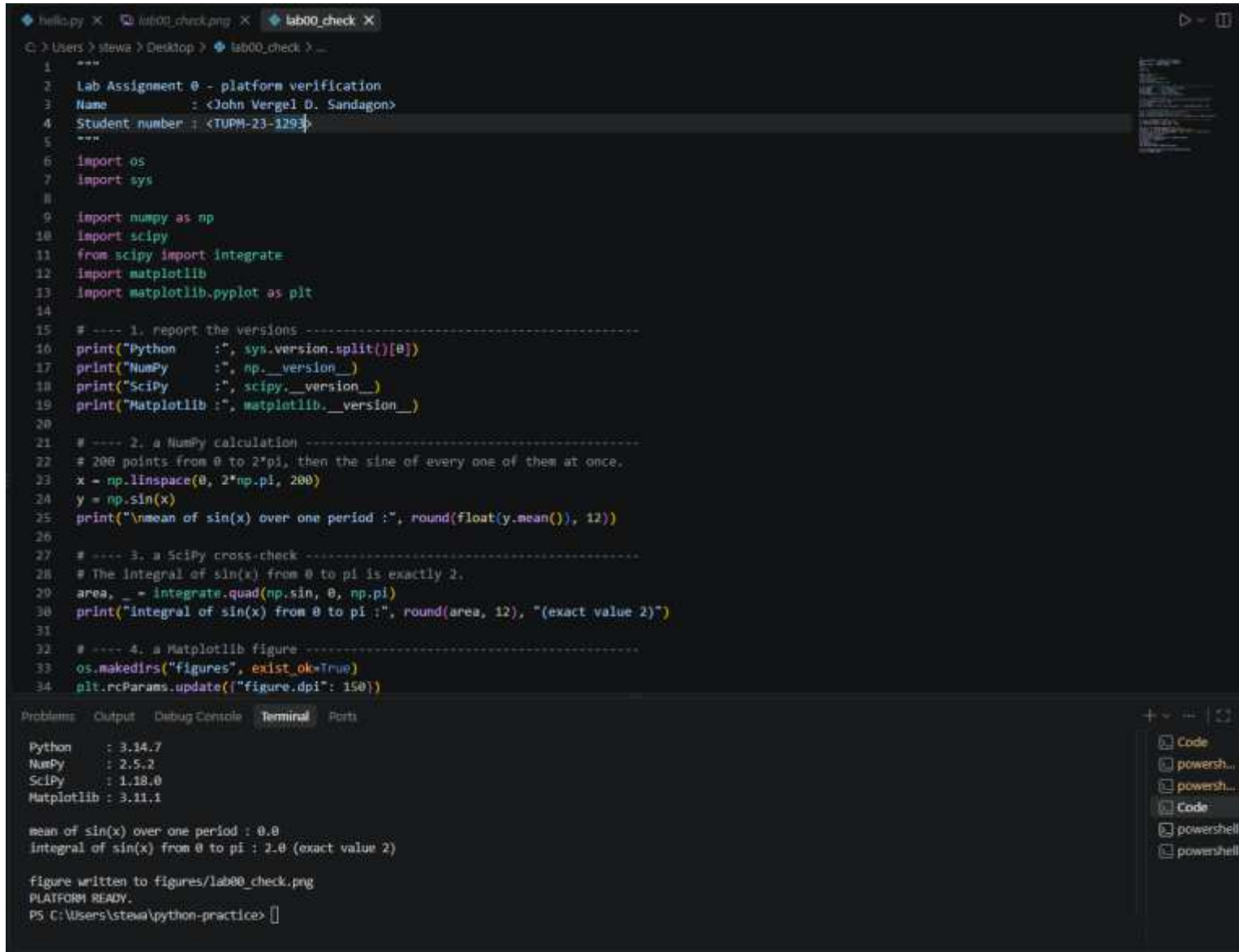


3. The terminal showing the successful installation of NumPy, SciPy and Matplotlib.

```
Problems Output Debug Console Terminal Ports

PS C:\Users\stewa\python-practice> python -m pip install --upgrade pip
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: pip in C:\Program Files\Python314\Lib\site-packages (26.2.1)
PS C:\Users\stewa\python-practice> python -m pip install numpy scipy matplotlib
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: numpy in C:\Users\stewa\AppData\Roaming\Python\Python314\site-packages (2.5.2)
Requirement already satisfied: scipy in C:\Users\stewa\AppData\Roaming\Python\Python314\site-packages (1.18.0)
Requirement already satisfied: matplotlib in C:\Users\stewa\AppData\Roaming\Python\Python314\site-packages (3.11.1)
Requirement already satisfied: contourpy>=1.0.1 in C:\Users\stewa\AppData\Roaming\Python\Python314\site-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in C:\Users\stewa\AppData\Roaming\Python\Python314\site-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.28.2 in C:\Users\stewa\AppData\Roaming\Python\Python314\site-packages (from matplotlib) (4.63.0)
Requirement already satisfied: kiwisolver>=1.3.1 in C:\Users\stewa\AppData\Roaming\Python\Python314\site-packages (from matplotlib) (1.5.0)
Requirement already satisfied: packaging>=20.0 in C:\Users\stewa\AppData\Roaming\Python\Python314\site-packages (from matplotlib) (26.3)
Requirement already satisfied: pillow>=9 in C:\Users\stewa\AppData\Roaming\Python\Python314\site-packages (from matplotlib) (12.3.0)
Requirement already satisfied: pyparsing>=3 in C:\Users\stewa\AppData\Roaming\Python\Python314\site-packages (from matplotlib) (3.3.2)
Requirement already satisfied: python-dateutil>=2.7 in C:\Users\stewa\AppData\Roaming\Python\Python314\site-packages (from matplotlib) (2.9.0.post0)
Requirement already satisfied: six>=1.5 in C:\Users\stewa\AppData\Roaming\Python\Python314\site-packages (from python-dateutil>=2.7->matplotlib) (1.17.0)
PS C:\Users\stewa\python-practice> █
```

4. The full console output of your verification program, ending in **PLATFORM READY.**



The screenshot shows a Visual Studio Code editor with a Python file named `lab00_check.py` open. The script is a verification program for a lab assignment. It includes comments for each step and prints out the results. The terminal window at the bottom shows the execution output, which matches the print statements in the code. The output ends with `PLATFORM READY.`

```
1  """
2  Lab Assignment 0 - platform verification
3  Name       : <John Vergel D. Sandagon>
4  Student number : <TUPM-23-1293>
5  """
6  import os
7  import sys
8
9  import numpy as np
10 import scipy
11 from scipy import integrate
12 import matplotlib
13 import matplotlib.pyplot as plt
14
15 # ---- 1. report the versions ----
16 print("Python      :", sys.version.split()[0])
17 print("NumPy       :", np.__version__)
18 print("SciPy        :", scipy.__version__)
19 print("Matplotlib    :", matplotlib.__version__)
20
21 # ---- 2. a NumPy calculation ----
22 # 200 points from 0 to 2*pi, then the sine of every one of them at once.
23 x = np.linspace(0, 2*np.pi, 200)
24 y = np.sin(x)
25 print("\nmean of sin(x) over one period :", round(float(y.mean()), 12))
26
27 # ---- 3. a SciPy cross-check ----
28 # The integral of sin(x) from 0 to pi is exactly 2.
29 area, _ = integrate.quad(np.sin, 0, np.pi)
30 print("integral of sin(x) from 0 to pi :", round(area, 12), "(exact value 2)")
31
32 # ---- 4. a Matplotlib figure ----
33 os.makedirs("figures", exist_ok=True)
34 plt.rcParams.update({"figure.dpi": 150})
```

Python : 3.14.7
NumPy : 2.5.2
SciPy : 1.18.0
Matplotlib : 3.11.1

mean of sin(x) over one period : 0.0
integral of sin(x) from 0 to pi : 2.0 (exact value 2)

figure written to figures/lab00_check.png
PLATFORM READY.
PS C:\Users\stewa\python-practice>

5. Your GitHub repository page, showing the repository name and the uploaded file.

The screenshot shows a GitHub repository page for 'numerical-methods-lab' owned by 'john-vergel-sandagon'. The repository is private and has 0 watchers, 0 forks, and 0 stars. The main branch is 'main'. The repository contains two files: 'README.md' (initial commit, 33 minutes ago) and 'hello.py' (Add files via upload, 33 minutes ago). The README file is selected and displays the text 'numerical-methods-lab'. The right sidebar shows sections for 'About' (no description), 'Releases' (no releases published), 'Packages' (no packages published), 'Contributors' (1 contributor: john-vergel-sandagon), 'Languages' (Python 100%), and 'Suggested workflows' (based on your tech stack).

john-vergel-sandagon / numerical-methods-lab

Code Issues Pull requests Agents Actions Projects Security and quality Insights Settings

numerical-methods-lab Private Watch 0 Fork 0 Star 0

main 1 Branch 0 Tags Go to file Add file Code

john-vergel-sandagon Add files via upload 10562e2 · 33 minutes ago 2 Commits

File	Commit	Time
README.md	Initial commit	33 minutes ago
hello.py	Add files via upload	33 minutes ago

README

numerical-methods-lab

About

No description, website, or topics provided

- Readme
- Activity
- 0 stars
- 0 watching
- 0 forks

Releases

No releases published

[Create a new release](#)

Packages

No packages published

[Publish your first package](#)

Contributors 1

john-vergel-sandagon Sandagon, J.

Languages

Python 100%

Suggested workflows

Based on your tech stack

figures/lab00_check.png

